

Emergence of Bistability in a Time-Delayed Stochastic miRNA–Protein Toggle Switch

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1 Abstract

This project models a miRNA-protein regulatory system with a delay in protein activation. The protein component is divided into an inactive (p_1) and active (p_2) form, with diffusion rate j controlling activation. Stochastic simulations implemented using the Gillespie algorithm show that for low j , the system stays in a single low-expression state. As j increases, the system begins to switch between low and high expression levels of p_2 and miRNA (m), indicating bistability. Furthermore, a theoretical rate-of-change analysis shows how increasing j effects the steady states.

2 Introduction

Normal cell growth is a tightly controlled biological process. Growth of a cell not only requires the ability of cells to proliferate, but also the ability to halt growth temporarily. This regulation is essential for various reasons; if cell growth is not properly controlled, it can lead to harmful outcomes such as proliferation of abnormal cells or uncontrolled overgrowth (Almalki, 2023). Both of these processes can lead to tumor development. It is thus necessary to stop the growth of cells at certain points so that we are able to prevent harmful processes from growing within the body.

One key regulator of this process is microRNA (miRNA), a biological complex found within the body. In the context of cell growth, miRNA functions as a temporary inhibitor, helping to suppress growth (Cai et al., 2009). This regulation is mediated through interactions with growth factors, which promote cellular proliferation. As growth factor proteins accumulate, they can then activate miRNA, which in turn inhibits the activity of these proteins. As a result, this reduces the ability of these growth factors to drive further cell growth. This interaction is commonly referred to as a negative feedback loop (Figure 1).

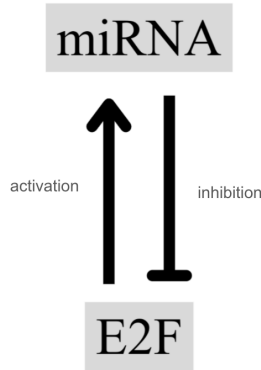


Figure 1: E2F, a common growth factor protein activates the miRNA. In turn, miRNA inhibits the growth factor protein such that it is unable to effect cells and help them grow.

From a mathematical perspective, this feedback mechanism between miRNA and growth factor protein is often modelled as a toggle switch, where the system transitions between states of high and low activity. Previously, Giampieri et al. (2011) had attempted to model this interaction between protein and miRNA using a system of ordinary differential equations (ODEs). Their model captured key processes such as production, degradation, and regulatory interactions of the protein and miRNA and is noted by Eq (1), describing the rate of change of protein (p) concentration, and Eq (2), describing the rate of change of miRNA (m) concentration:

$$\dot{p} = a + \frac{k_1 p^2}{G_1 + p^2 + G_2 m} - dp \quad (1)$$

$$\dot{m} = b + k_2 p - \gamma m \quad (2)$$

For the first equation (Eq 1) the term a represents the basal production rate of the protein, corresponding to constant background synthesis. The nonlinear term

$$\frac{k_1 p^2}{G_1 + p^2 + G_2 m}$$

represents the activation of the protein. The p^2 term in the numerator represents positive feedback, where higher protein levels enhance their own production. The denominator introduces saturation and inhibition effects: G_1 is a baseline saturation constant, the p^2 term prevents unbounded growth, and $G_2 m$ represents inhibition by miRNA. The term $-dp$ corresponds to degradation of the protein.

In the second equation (Eq 2), the term b represents basal miRNA production. The term $k_2 p$ models activation of miRNA by the protein, indicating that higher protein levels promote miRNA production. The degradation of miRNA is captured by the term $-\gamma m$.

Within Giampieri’s system, the term for protein activation is embedded within the ODE for the protein p alongside protein production. One key assumption that Giampieri is thus making in his model is that protein activation is instantaneous. Biologically, this assumption is not realistic. In most cases, newly made proteins within the body are not active. They must often go through further modifications and activation processes in order to achieve an active form (Alsallaq & Zhou, 2007). Additionally, depending on cellular conditions, the protein may not become active for longer periods of time, so it activates at a slower rate in comparison to more active systems.

In this project, we address this assumption by introducing a modification to the existing ODE system framework. Rather than assuming instantaneous activation or imposing a fixed delay through delay differential equations (DDEs), we model activation of the protein as a rate-limited conversion from inactive to active protein. While a DDE formulation could be used, it is not suitable as our focus is on simulating this system stochastically. Using a DDE would make the system non-Markovian, where the transition probabilities depend on the history of the system. In this case, we would not be able to use Gillespie’s algorithm.

We want to be able to use Gillespie’s algorithm because switching behaviour in regulatory systems such as the miRNA–protein toggle switch is often driven by intrinsic noise rather than deterministic behaviour. While the ODE model describes average behaviour, it cannot capture random transitions between states. Gillespie’s algorithm assumes a Markovian system, in which reaction propensities depend only on the current state and not on the history of the system (Gillespie, 1977). Thus, to retain the Markov property, we instead introduce new terms to effect activation rather than adding an explicit delay.

Our primary goal is to investigate how non-instantaneous protein activation influences the behaviour of the miRNA–protein toggle switch. In particular, we want to examine how varying the activation rate affects the system’s switching dynamics compared to the original instantaneous activation model.

3 Methodology

3.1 Model Modification

We began by modifying the original miRNA–protein toggle switch model to account for non-instantaneous activation of the protein. In the original system, the protein p is assumed to become active immediately upon production. We split this protein into two components: inactive protein p_1 and active protein p_2 . The

modified system is given by

$$\dot{p}_1 = a - dp_1 - jp_1 \quad (3)$$

$$\dot{p}_2 = \frac{k_1(p_2)^2}{G_1 + (p_2)^2 + G_2m} - dp_2 + jp_1 \quad (4)$$

$$\dot{m} = b + k_2p_2 - \gamma m \quad (5)$$

Here, the term jp_1 has been introduced to describe conversion from inactive to active protein, where j is the activation rate parameter. This means varying j allows us to control the timescale of activation and study how this affects the overall system dynamics.

According to Eq. 3, the inactive protein p_1 is only being produced and degraded; in order to become active and interact with miRNAs, it must be in the active p_2 form. Only p_2 participates in the feedback loop with miRNA as noted by its ODE holding the activation term; it also promotes miRNA production through the k_2 term. Thus, p_1 acts as a reservoir of inactive protein, while p_2 is the biologically functional species which drives regulation.

3.2 Gillespie's Algorithm

To study stochastic switching behaviour of this model, we implemented both the unmodified and modified systems using Gillespie's algorithm.

A standard approach to simulating stochastic systems is to iterate the model forward in fixed time steps. However, to accurately capture every individual reaction event, these time steps must be very small. This is very expensive computationally because many iterations are required, and most steps result in no reactions occurring, and thus no change in our system. Gillespie's algorithm overcomes this by directly simulating reaction events. Rather than stepping through time at fixed intervals, the algorithm automatically samples the time to the next reaction and selects which reaction will occur based on the current rates in the system (Gillespie, 1977). In this way, the simulation progresses by jumping between events, avoiding unnecessary intermediate steps.

For the modified and unmodified systems, these rates are defined as follows:

Unmodified System

Reaction	Rate	Effect
Protein production	a	increases p
Protein degradation	dp	decreases p
Protein self-activation	$\frac{k_1p^2}{G_1 + p^2 + G_2m}$	increases p
miRNA production	b	increases m
miRNA degradation	γm	decreases m
miRNA activation by protein	k_2p	increases m

Table 1: The rates and effects of the unmodified system as defined by equations (1) and (2).

Modified System

Reaction	Rate	Effect
Inactive protein production	a	increases p_1
Inactive protein degradation	dp_1	decreases p_1
Conversion ($p_1 \rightarrow p_2$)	jp_1	decreases p_1 , increases p_2
Active protein self-activation	$\frac{k_1 p_2^2}{G_1 + p_2^2 + G_2 m}$	increases p_2
Active protein degradation	dp_2	decreases p_2
miRNA production	b	increases m
miRNA degradation	γm	decreases m
miRNA activation by active protein	$k_2 p_2$	increases m

Table 2: The rates and effects of the modified system as defined by equations (3), (4), and (5).

Note that for both systems, the molecule count of p or m will be increase or decrease by 1 for a rate occuring at a given point in time.

3.3 Gillespie's Algorithm Implementation

At each step of the algorithm, we let the reaction rates be denoted by $r_1, \dots, r_n$. The total reaction rate of the system is then given by

$$\lambda = \sum_{i=1}^n r_i, \quad (6)$$

which represents the overall propensity for any reaction to occur. We have 6 such reaction rates for the unmodified system (Table 1) and 8 such rates for the modified system (Table 2).

Instead of specifying a uniform time step, the waiting time until the next reaction, τ , is sampled from an exponential distribution with parameter λ . Using inverse transform sampling, we draw u and calculate τ as such:

$$u \sim \text{Uniform}(0, 1), \quad (7)$$

$$\tau = -\frac{\ln(u)}{\lambda}. \quad (8)$$

Calculating the next time step as such ensures that reaction events occur at random time intervals consistent with the total reaction rate, capturing the stochastic timing inherent in the system.

For each r_i , a frequency for each reaction is calculated as follows:

$$\mathbb{P}(\text{reaction } i) = \frac{r_i}{\lambda},$$

the reaction with the highest frequency is the one that proceeds in the system at a given point in time. Molecule counts are updated according to the effects listed for the rate.

Using this algorithm, we can start at time point 0 and specify a maximum time point to run our system until. This means the system will begin updating molecule counts at time 0 and will only stop updating molecule counts once it reaches the maximum specified time point.

Parameters

All values except for p and m are constants, as p and m are updated at every time step and thus depend on their most recent values. We use the constants provided by Giampieri et al. (2011). Note that Giampieri does not provide a value for j as we introduced this into our modified system; our focus is on observing what

happens when we modify this value of j , so we compare two cases where $j = 0.1$ (lower activation case) and $j = 0.4$ (higher activation case).

For our unmodified system, we specified both our initial values of p and m to be 0. For our modified system, we specified our initial values of p_1 , p_2 , and m to be 100, 50, and 30, respectively.

Both systems were run to a maximum time step of 100,000. This ensured that sufficient data was collected to construct reliable time series and histograms.

3.4 Histogram Construction

To analyze the distribution of protein values obtained from running Gillespie’s algorithm on both systems, histograms were constructed from the time series output. Since Gillespie’s algorithm alters molecule counts at irregular time intervals, the raw output cannot be directly grouped into histogram bins, as this would not accurately represent how long the system remains at each protein level. In particular, simply counting the observed values and grouping them into set intervals would place more weight on periods where reactions occurred more frequently, rather than the actual time spent in a given state.

To account for this, we first used the Gillespie output to construct a piecewise constant function of molecule count over time. Between reaction events, the molecule count remains unchanged, so the function stays constant until the next event occurs. We then sampled this function at evenly spaced time intervals, so that each sampled value corresponded to the same amount of elapsed time. This ensured that the histogram reflected the amount of time the system spent in different expression ranges, rather than the frequency of reaction events.

After obtaining uniformly sampled values, fixed-width bins were used to group the observed molecule counts. Bins were constructed starting from 0 and extended to the maximum observed value in the simulation to ensure the full range of data was captured. A bin width of 1 was used to preserve the full resolution of molecule counts. While larger bin widths could also be used, they would group together nearby protein levels and smooth out important lower and higher states of the distribution for analysis.

3.5 Stability Analysis

To complement the stochastic simulations, we performed a stability analysis of the deterministic modified system.

From setting Eq. 3 to 0 as such

$$0 = a - dp_1 - jp_1,$$

we obtain

$$p_1 = \frac{a}{d+j}.$$

Similarly, from setting Eq. 5 to 0, as such

$$0 = b + k_2p_2 - \gamma m,$$

we obtain

$$m = \frac{b + k_2p_2}{\gamma}.$$

Substituting these expressions into the p_2 equation reduces the system to a single equation in p_2 :

$$\frac{dp_2}{dt} = \frac{k_1p_2^2}{G_5 + p_2^2 + G_4p_2} - dp_2 + j\left(\frac{a}{d+j}\right).$$

Here, the constants G_4 and G_5 arise from substituting the expression for m into the denominator. These values are as follows:

$$G_4 = \frac{G_2k_2}{\gamma}$$

$$G_5 = G_1 + G_3$$

The equilibrium values of p_2 are determined by the roots of the right-hand side of the reduced equation. To identify these, we plotted the function

$$f(p_2) = \frac{k_1 p_2^2}{G_5 + p_2^2 + G_4 p_2} - d p_2 + j \left(\frac{a}{d + j} \right) \quad (9)$$

against a horizontal line 0.

To determine stability, we analyzed the sign of $f(p_2)$ around each root. If $f(p_2)$ changes from positive to negative as p_2 increases through a root, the equilibrium is stable. If $f(p_2)$ changes from negative to positive, the equilibrium is unstable.

4 Results

4.1 Unmodified System

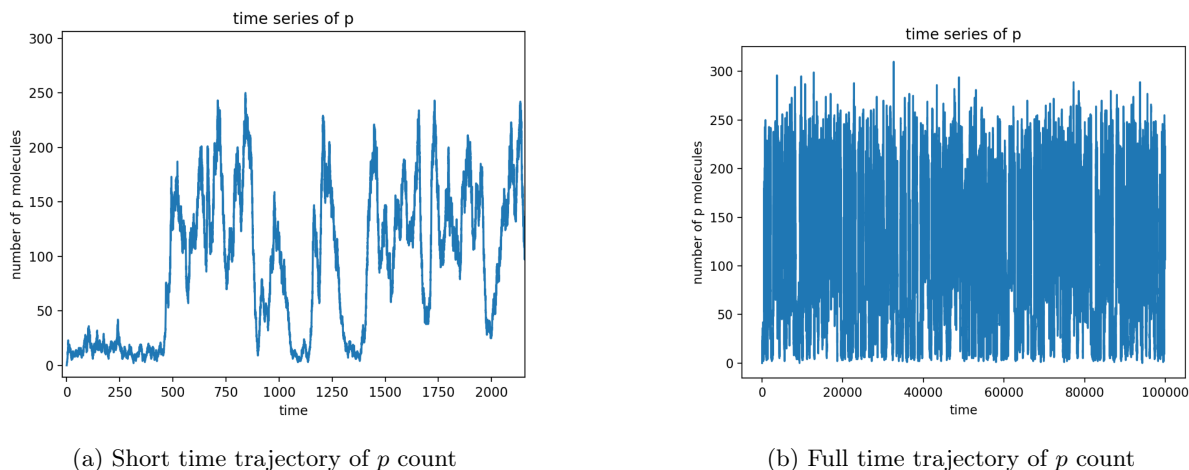


Figure 2: Time series of protein concentration p in the unmodified system. (a) Short-time simulation showing initial transient behaviour and beginning of consistent transitions between low and high expression states. (b) Longer-time simulation demonstrating sustained switching between states over extended time.

To evaluate the behaviour of the unmodified miRNA-protein toggle switch model, stochastic simulations were performed using Gillespie's algorithm. The time series of protein concentration p was examined over both short and full length timescales (Figure 2).

In the short-time simulation (Figure 2a), the system initially remains within a low-expression state, with p values fluctuating at relatively small magnitudes. As time progresses, the system begins to transition between both lower and higher protein levels. The longer-time simulation (Figure 2b) shows that this behaviour persists as p continues to fluctuate across this general range of values over extended periods of time. The system repeatedly moves between low and high protein levels, with no prolonged confinement to a single region. This behaviour indicates that the system does not converge to a single expression level, but instead exhibits sustained transitions between low and high protein states.

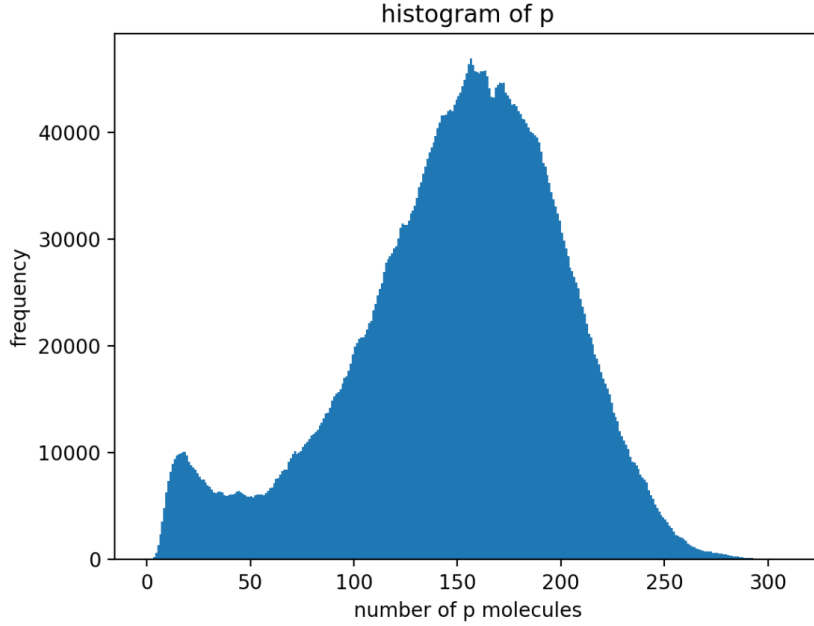


Figure 3: Histogram of protein concentration p in the unmodified system. Distribution shows peaks at both low and high p values, indicating that system occupies multiple ranges of expression. Differences in peak height reflect variation in the amount of time spent in each range of p .

To evaluate how frequently different protein levels occur in the unmodified system, a histogram of p was constructed over the full simulation (Figure 3). Unlike the time series, which shows how p changes over time, the histogram reflects how long the system resides within different ranges of p .

As shown in Figure 3, the distribution of p exhibits two distinct peaks: a smaller peak at lower protein values and a larger peak at higher protein values. This indicates that the system occupies both low and high expression regions, but does not spend equal time in each. The difference in peak heights reflects the relative amount of time the system spends in each region. The smaller peak at low p values suggests that the system transitions through this region relatively quickly, whereas the larger peak at higher p values indicates that the system remains in this region for longer periods of time.

Together, these results suggest that while the system is capable of accessing both low and high expression states, it more frequently resides in the high p region. This asymmetry suggests that the protein quickly transitions into the higher, active state and prefers to stay at this state as well.

4.2 Modified System

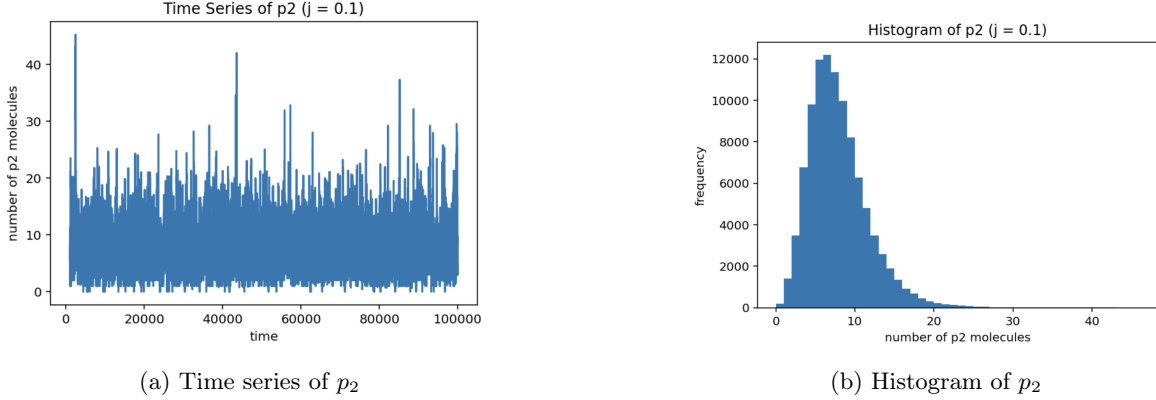


Figure 4: Time series and histogram of active protein concentration p_2 in the modified system for $j = 0.1$. (a) Time series showing that p_2 remains within a low-expression range with only occasional, short-lived increases. (b) Histogram showing a strong concentration of values within the low p_2 range, with a diminishing tail toward higher values, indicating that higher expression levels are not sustained.

To evaluate how introducing an activation rate affects system behaviour, stochastic simulations were performed for the modified system with parameter $j = 0.1$. The time series and corresponding histogram of the active protein concentration p_2 are shown in Figure 3.

In the time series (Figure 4a), p_2 remains largely confined to a low-expression range throughout the entire simulation. While occasional spikes to higher values are observed, these increases are brief and not sustained over time. The system quickly returns to lower p_2 values following these fluctuations, indicating that higher expression levels are not stable.

The histogram (Figure 4b) supports this observation, showing that the majority of values are concentrated within a low p_2 range (approximately 0 to 20 molecules). Although there is a tail extending toward higher values, there is no distinct peak in this region, and the frequency of larger p_2 values fades off rapidly.

Together, these results indicate that the system predominantly resides in a low-expression state when $j = 0.1$. The lack of a secondary peak and the absence of sustained high p_2 values suggest that switching behaviour observed in the unmodified system is suppressed under low activation rates, resulting in a single dominant expression regime.

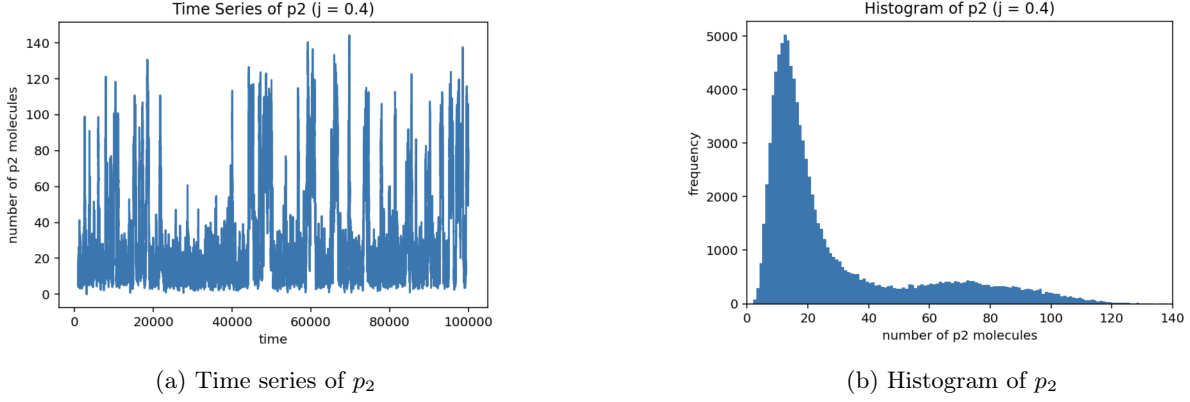


Figure 5: Time series and histogram of active protein concentration p_2 in the modified system for $j = 0.4$. (a) Time series showing sustained fluctuations across both low and high expression ranges. (b) Histogram showing a primary peak at low p_2 values and a secondary distribution extending into higher values, indicating increasing occupancy of higher expression regions.

To evaluate how increasing the activation rate affects system behaviour, stochastic simulations were performed for the modified system now with $j = 0.4$. The time series and corresponding histogram of the active protein concentration p_2 are shown in Figure 5.

In the time series (Figure 5a), p_2 exhibits sustained fluctuations across a broader range of values compared to the $j = 0.1$ case (Figure 4a). In contrast to the low activation regime, higher p_2 values are no longer rare or short-lived, but instead occur more frequently and are maintained for longer periods of time. This indicates that increasing j allows the system to access and persist in higher expression regions.

The histogram (Figure 5b) supports this observation. While a primary, dominant peak remains at lower p_2 values, there is a clear extension of the distribution into higher ranges, with increased frequency of intermediate and larger p_2 values compared to the $j = 0.1$ case. This suggests that the system spends more time in these higher expression regions, rather than remaining confined to the low-expression state.

Together, these results indicate that increasing the activation rate j alters the system behaviour by enabling sustained occupation of higher p_2 values. As j increases, faster conversion from p_1 to p_2 leads to greater availability of active protein, allowing the system to maintain higher expression levels and exhibit persistent switching across multiple regions of expression.

4.3 Stability Analysis for Modified System Cases

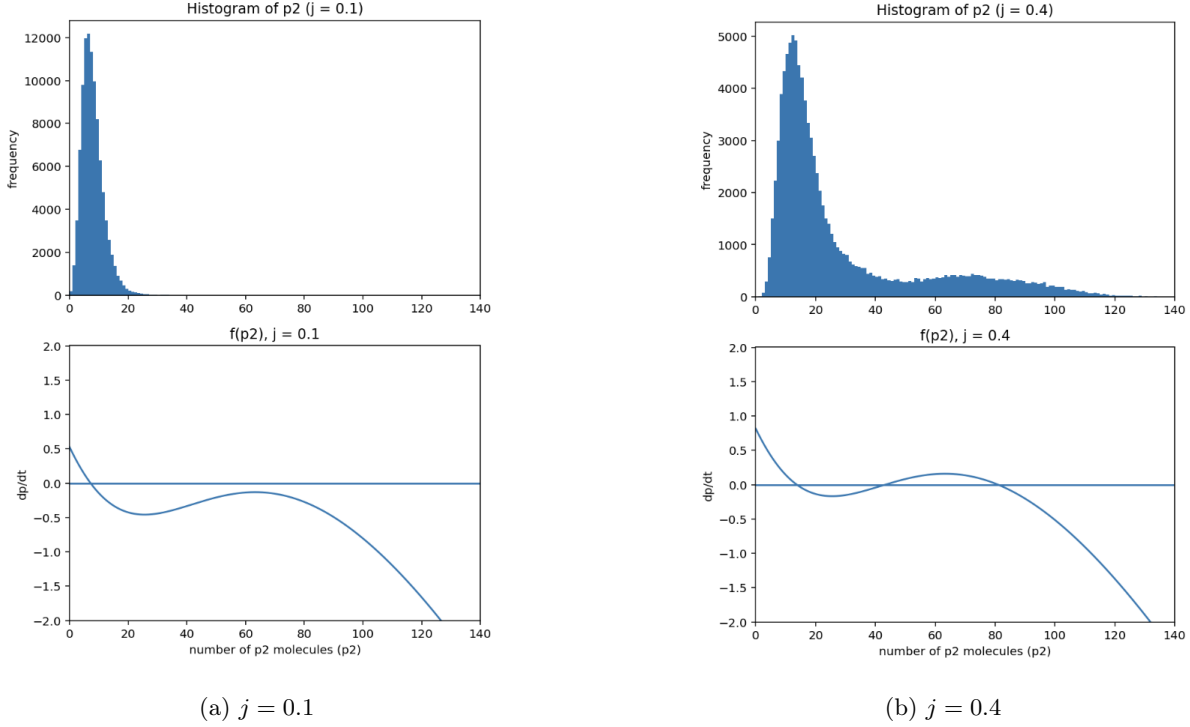


Figure 6: Stability analysis of the modified system for $j = 0.1$ (a) and $j = 0.4$ (b). Histograms of p_2 are shown above, with corresponding plots of $f(p_2)$ below. Intersections of $f(p_2)$ with the horizontal axis indicate equilibrium points. (a) A single stable equilibrium is observed. (b) Two stable equilibria separated by an unstable equilibrium are present.

To understand the underlying structure of the modified system, stability analysis was performed. The corresponding histograms and plots of $f(p_2)$ for $j = 0.1$ and $j = 0.4$ (Eq. 9) are shown in Figure 5. Note that the histograms are the same as those presented in Figure 4b and 5b.

For $j = 0.1$, the function $f(p_2)$ has a single intersection with the horizontal axis (Figure 6a). Analysis of the sign of $f(p_2)$ around this point shows that it is stable, indicating that the system has one steady state. This implies that, regardless of initial conditions, the system is driven toward the same p_2 value. The system thus remains localized within this region and does not sustain itself at higher expression levels.

In contrast, for $j = 0.4$, the function $f(p_2)$ exhibits three intersections with the horizontal axis (Figure 6b). Of these, only two correspond to stable equilibrium points, while the middle point is unstable. This structure indicates that the system has two stable states separated by an unstable boundary. As a result, the system can settle into either a low or high p_2 region depending on its current state, leading to bistable behaviour. the system would not settle, and is shown to have low frequencies, around the unstable middle point.

These results are consistent with the histograms of molecule frequencies observed for the stochastic simulations. For $j = 0.1$, the presence of a single stable equilibrium explains why the system remains confined to a low-expression region. For $j = 0.4$, the existence of two stable points explains the sustained switching and occupation of both low and high expression ranges observed in the time series and histograms. Thus, the stability analysis provides a theoretical explanation for the behaviour observed in the stochastic simulations.

5 Discussion

Results from this project show that introducing a rate-limited activation step significantly alters the behaviour of the miRNA-protein toggle switch. In particular, increasing the activation rate j enhances the conversion from inactive protein p_1 to active protein p_2 , increasing the availability of active protein and thus strengthening the feedback interactions within the system.

At low values of j , $j = 0.1$, the feedback is not strong enough to sustain activation of the protein. As a result, the system collapses into a single stable state, where p_2 remains confined to a low-expression range. While we do observe that some fluctuations to higher values are present, these events are rare and not sustained. In contrast, at higher values of j , $j = 0.4$, the increased rate of conversion allows sufficient active protein to accumulate, strengthening the feedback loop between the miRNA and protein. This enables the system to support two self-sustaining expression states, where p_2 can remain in either a low or high range depending on its current state, showcasing sustained and relatively frequent switching behaviour in comparison.

However, although increasing j from 0.1 to 0.4 restores switching behaviour, the extent of switching remains reduced compared to the unmodified system, where transitions between expression states occur more frequently and across a broader range of values. In comparison to both cases of the modified system, the protein in the unmodified system typically remains in a higher expression state. This observation aligns with the instantaneous activation behaviour we noted with the unmodified system.

The stability analysis provides a theoretical framework for understanding these observations by identifying the equilibrium states that the system is more driven towards. However, a limitation of this analysis is that it is based on a theoretical curve, and does not directly quantify how closely the stochastic simulations follow these predicted equilibrium values. An area for future work would be incorporating methods to compare the simulated distributions with the theoretical predictions would strengthen this connection.

Overall, this project demonstrates that introducing non-instantaneous activation can significantly impact switching behaviour in the miRNA-protein toggle switch model. Importantly, this effect depends strongly on the value of the activation rate j . At sufficiently low values of j , switching behaviour is suppressed, whereas at higher values, switching emerges due to the presence of multiple stable states. However, the disappearance of switching at low j may reflect a scenario in which activation is relatively slow compared to other processes in the system, rather than being solely a consequence of introducing delay. This suggests that the relationship between activation rate and switching behaviour is highly sensitive to parameters we choose to analyze. Future work should therefore focus on identifying biologically realistic ranges for j and other system parameters, to determine whether the observed behaviours are consistent with real biological systems.

6 References

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